

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF SOFTWARE ENGINEERING BATCH(2025)

FESE-SPRING SEMESTER 2025 MID TERM TEST

TIME:90 MINS

DATED:28-2-2025

APPLIED PHYSICS PH-122

ATTEMPT ALL QUESTIONS

MAX.MARKS:20

Q1.		CLO1
a)	Discuss graphical techniques and advantages of converting nonlinear curve into linear curve?	3
b)	What is scalar Triple product and vector Triple product show that: $\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C}) \vec{B} - (\vec{A} \cdot \vec{B}) \vec{C}$	4
c)	i) Define Gradient, Divergence, Curl? ii) Explain line integral of vector.	3
Q.2		CLO2
a)	If $\{F=(2xy+z^3)I + (x^2j) + 3xz^2k\}$ where F is a force acting on a body i) Show that it is conservative force. ii) There exist a scalar function ϕ such that $F=\nabla \phi$.	2.5
b)	Find grad ϕ , if $\phi = \ln r $	2.5
c)	If $A = (3x^2 + 6y)I - 14yzj + 20xz^2k$ evaluate $\int A \cdot dr$ from (0,0,0) to (1,1,1) following the path $x=t, y=t^2, z=t^3$.	2.5
d	SHOW that $A=(2x^2 + 8xy^2z)I + (3x^3y - 3xyj - (4y^2z^2 + 2x^3z)k$ is not solenoidal but $B=xyz^2$ (A) is solenoidal	2.5

$$\frac{1}{2} - 1 = -\frac{1}{2}$$

$$= -\frac{1-2}{2}$$

$$= -\frac{1}{2}$$